

RPI_Rcode

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AMR Gene Risk Scoring and Resistance Persistence Index (RPI) Calculation Workflow

Use reshaped dataset (`combined_amr_results_seasons_reshaped.csv`) when applying RPI. Reshaped dataset can be found in thesis GitHub repository.

R code asses RPI by evaluating ARG seasonal stability through the persistence index, applying Zhang's risk rank assessment by evaluating mobility (present in plasmid), ESKAPE pathogen association, and enviroment (human-associated enviroment). Zhang risk rank is converted to Zhang risk score.

Zhang risk score and persistence index are added together to get RPI.

```
# Load required libraries
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(stringr)
library(tidyr)

# Import data
df <- read.csv("/Users/michael/Documents/Masters/MSThesis/RCode/RPI/combined_amr_results_seasons.csv")

# Clean data: remove missing, blank, or ambiguous species entries
df <- df %>%
  filter(
```

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    !is.na(read_species),
    read_species != "",
    !str_detect(read_species, "\\(chromosome or plasmid\\)")
  )

# Add column indicating human association (all entries are human-associated)
df <- df %>%
  mutate(Human_Associated = "Yes")

# Identify ESKAPE pathogens
eskape_species <- c(
  "Enterococcus faecium", "Staphylococcus aureus", "Klebsiella pneumoniae",
  "Acinetobacter baumannii", "Pseudomonas aeruginosa", "Enterobacter"
)

df <- df %>%
  mutate(ESKAPE = if_else(
    str_detect(read_species, str_c(eskape_species, collapse = "|")),
    "Yes", "No"
  ))

# Count plasmid and chromosome mentions in each row
df <- df %>%
  mutate(
    plasmid_count = str_count(tolower(read_species), "plasmid"),
    chromosome_count = str_count(tolower(read_species), "chromosome")
  )

# Combine entries by gene and season
df_combined <- df %>%
  group_by(gene_name, Season) %>%
  summarise(
    rpm = sum(rpm, na.rm = TRUE),
    plasmid_count = sum(plasmid_count, na.rm = TRUE),
    chromosome_count = sum(chromosome_count, na.rm = TRUE),
    Human_Associated = first(Human_Associated),
    ESKAPE = first(ESKAPE),
    .groups = "drop"
  ) %>%
  mutate(
    Mobility_Score = plasmid_count / (plasmid_count + chromosome_count)
  )

# Calculate persistence score: 0 if only present in one season, else scaled by seasonal balance
rpm_summary <- df_combined %>%
  group_by(gene_name) %>%
  summarise(
    Winter = sum(rpm[Season == "Winter"], na.rm = TRUE),
    Summer = sum(rpm[Season == "Summer"], na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    Persistence_Score = case_when(

```

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    Winter == 0 | Summer == 0 ~ 0,
    TRUE ~ 1 - abs(Winter - Summer) / (Winter + Summer + 1e-6)
  )
)

# Join persistence score to combined table
df_combined <- df_combined %>%
  left_join(rpm_summary %>% select(gene_name, Persistence_Score), by = "gene_name")

# Assign Zhang Score based on risk factors
df_combined <- df_combined %>%
  mutate(Zhang_Score = case_when(
    Human_Associated == "Yes" & Mobility_Score > 0 & ESKAPE == "Yes" ~ 4,
    Human_Associated == "Yes" & Mobility_Score > 0 & ESKAPE == "No" ~ 3,
    Human_Associated == "Yes" & Mobility_Score == 0 ~ 2,
    TRUE ~ 1
  ))

# Invert Zhang Score to rank (lower = higher risk)
df_combined <- df_combined %>%
  mutate(Zhang_Rank = case_when(
    Zhang_Score == 4 ~ 1,
    Zhang_Score == 3 ~ 2,
    Zhang_Score == 2 ~ 3,
    Zhang_Score == 1 ~ 4,
    TRUE ~ NA_real_
  ))

# Calculate Resistance Persistence Index (RPI)
df_combined <- df_combined %>%
  mutate(RPI_Score = Persistence_Score + Zhang_Score)

# Export cleaned and scored dataset
write.csv(df_combined, "combined_amr_results_seasons_with_RPI.csv", row.names = FALSE)

```